LESSON 3.4
DIVIDING RATIONAL NUMBERS

LESSON INTRODUCTION

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

LEARNING TARGETS

- I can fluently divide rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How does the process of dividing integers apply to dividing rational numbers?
- How do I divide positive and negative rational numbers?

LESSON NARRATIVE

This lesson builds on the integer rules for division as students review the rules and apply them to the division rational numbers. A brief review of the division of positive rational numbers and the sign rules for the division of integers are used to help students determine the sign rules for the division of rational numbers. Students apply the sign rules to divide rational numbers.

COMPONENT SUMMARY

3.4.1 WARM-UP *(~5 MIN.)*

Determine the value of a stack of quarters

3.4.2 REFRESHER *(10 MIN.)*

Review the division of positive rational numbers

3.4.3 COLLABORATION *(10 MIN.)*

Review sign rules for the division of integers

3.4.4 GUIDED PRACTICE *(10 MIN.)*

Apply the sign rules to the division of positive and negative rational numbers

3.4.5 YOUR TURN! *(10 MIN.)*

Practice division of positive and negative rational numbers

3.4.6 WRAP-UP *(~5 MIN.)*

Lesson reflection

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (optional) Calculators (to check work)

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may change the divisor to its opposite rather than the reciprocal when dividing.
- Students may incorrectly change an improper fraction to a mixed number, forgetting that any remainders need to be written as a fraction with the given denominator.

THINGS TO CONSIDER

- If students ask why the rules for multiplying and dividing signed numbers are the same it may be helpful for them to be reminded that division such as $\frac{2}{3} \div -4$ is the same as $\frac{2}{3} \cdot -\frac{1}{4}$.
- Students should only use a calculator for the verification in 3.4.3 Collaboration. Consider how to monitor your students. Options could be that they do not receive a calculator until they can show you, they did the work or to come back as a whole group for the verification.
- Determine if your students need the support of all or part of the refresher.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

3.4.4 Guided Practice positions students to apply the sign rules for dividing rational numbers without actually solving the given expressions. Consider applying “Notice and Wonder” to give students who may still be struggling with sign rules an opportunity to talk through the task to solidify their thinking before moving on to 3.4.5 Your Turn!

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students will be applying the sign rules for division to divide rational numbers fluently. Clarifications within this standard emphasize “Adapt(ing) procedures to apply them to a new context” – utilize the progression of the lesson from reviewing the division of positive rational numbers and the rules for dividing integers to bringing them together to determine how to divide positive and negative rational numbers in 3.4.4 Guided Practice.

DIFFERENTIATE INSTRUCTION

The lesson’s objective is for students to divide rational numbers fluently. In 3.4.3 Collaboration, partners work together to review the sign rules for the division of integers. Both are brought together in 3.4.4 Guided Practice, where students divide positive and negative rational numbers. Consider using visual models for students who are still struggling to remember the rules and procedures. A three-column chart can be used to record the sign rules for division, and number lines or other visual models can help students review the division of integers and to show the division of an integer and a rational number.

Dividend	Divisor	Quotient

3.4.1 WARM UP: WOULD YOU RATHER?

ENRICHMENT

Consider engaging prior knowledge on division byprompting students with these questions:

- If a quarter is about 0.069 of an inch thick, how can you use the thickness to estimate which stack you would rather have?
- Which operation makes the most sense to help with estimation? Why?

3.4.1 Warm-Up: Would You Rather?

1. Explain if you would rather have a stack of quarters that stands from the floor to the top of your head **OR** $225.00.

   Answers will vary. I would rather have $225.00, if I am 62 inches or less. The width of a quarter is 0.069 inches.

   $\frac{62}{0.069} \approx 898.55 \cdot 0.25 = 224.64$

   $224.64

2. Explain whether you would change your mind if the quarters were stacked to the top of your desk.

   Answers will vary. No, I am taller than my desk, so I would still rather have $225.00.

3.4.2 Refresher: Dividing Positive Rational Numbers

1. Ren found the following quotient, making at least one error in her work.
   a. Circle the error(s) in Ren's work and write an explanation of the error(s). Then, find the correct quotient.

Ren's Work	Explanation	Correct Quotient
$12.636 \div 3.12$ $\begin{array}{r} .040 \\ 3.12\overline{)12.636} \\ -12\,48 \\ \hline 156 \\ -\ 0 \\ \hline 0 \end{array}$	*Ren should have moved the decimal point two times in the divisor and dividend – making the divisor a whole number.*	$12.636 \div 3.12$ $\begin{array}{r} 4.05 \\ 312\overline{)1263.6} \\ -1248 \\ \hline 1560 \\ -\ 1560 \\ \hline 0 \end{array}$

   b. What advice do you have for Ren that could help her with dividing decimals in the future?
   Answers will vary. One thing to remember when making your divisor a whole number is moving the decimal point in the dividend the same number of times as the divisor.

2. Trevor found the following quotient, making at least one error in his work. Circle the error(s) in Trevor's work. Write an explanation of the error(s). Then, find the correct quotient.

Trevor's Work	Explanation	Correct Quotient
$\left(\frac{38}{13}\right) \div \left(1\frac{6}{13}\right)$ $\left(\frac{38}{13}\right) \div \left(\frac{19}{13}\right)$ $\frac{(38 \div 19)}{13}$ $\frac{2}{13}$	*Answers will vary. Trevor should have applied the multiplicative inverse to find quotient.*	$\frac{38}{13} \div \frac{19}{13}$ $\downarrow$ $\frac{38}{13} \cdot \frac{13}{19} = \frac{494}{247} = 2$

3.4.2 REFRESHER: DIVIDING POSITIVE RATIONAL NUMBERS

TEACHER MOVES

Encourage students to use precise mathematical vocabulary by prompting them to use the terms 'divisor' and 'dividend' in their response to question 1b.

DIFFERENTIATE INSTRUCTION

Some students may struggle with keeping the numbers and decimal points in alignment when using long division.Consider having them rewrite and solve the question using grid paper to work it out, and then copy it into the workbook or cutting the grid paper to fit and glue or tape into the workbook.

TEACHER MOVES

Students may mention that Trevor should have multiplied by the reciprocal. Consider asking students to clarify their answers:

•The reciprocal of what -- the dividend, the divisor, or both?

•What property of equality is used when multiplying by the reciprocal?

3.4.3 COLLABORATION: SIGNED NUMBER RULES

DIFFERENTIATE INSTRUCTION

Some students may struggle to keep track of the sign rules. Consider having them build a three-column chart first, with the labels 'Dividend.' 'Divisor,' and 'Quotient'. Have them fill in the columns with either the signs only or using actual positive and negative numbers to summarize the four rules.

TEACHER MOVES

Students should only use a calculator for the verification of the answer so that students can formally state the rule.

Consider asking students if the rules also apply if the forms of the rational numbers differ as in -4.25 ÷ $\frac{1}{3}$.

DIFFERENTIATE INSTRUCTION

Have students write and solve their own expressions where they are dividing a fraction by a decimal and a decimal by a fraction. Challenge them further by asking them to write one of the fractions with a decimal numerator and denominator.

3.4.3 Collaboration: Signed Number Rules

1. Fill in the blanks with the words *positive* or *negative* to write the rules for dividing integers.
   - The quotient of two positive integers is a _*positive*_ integer.
   - The quotient of two negative integers is a _*positive*_ integer.
   - The quotient of a negative integer divided by a positive integer is a _*negative*_ integer.
   - The quotient of a positive integer divided by a negative integer is a _*negative*_ integer.
2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.
   Answers will vary. The rules for dividing rational numbers is the same as the rules for dividing integers. This is true because division can be rewritten as multiplication by the multiplicative inverse of the divisor.
3. Complete the table. Use a calculator to find the quotient of each expression to verify if the same rules for signs apply when dividing rational numbers.

Expression	Sign of Quotient Using the Rules	Quotient	Did the Same Rule Apply?
$\left(-2\frac{3}{8}\right) \div \left(-1\frac{1}{4}\right)$	*Positive*	$1\frac{9}{10}$	*Yes*
$6\frac{7}{18} \div -4\frac{2}{9}$	*Negative*	$-1\frac{39}{76}$	*Yes*
$43.008 \div -4.096$	*Negative*	-10.5	*Yes*
$(-525.6) \div (-0.36)$	*Positive*	1,460	*Yes*

4. Complete the sentence.
   The rules for dividing rational numbers are

   ○ the same as
   ○ different from

   the integer rules for division.

3.4.4 Guided Practice: Finding the Quotients of Rational Numbers

1. Circle the expressions that will result in a positive quotient.

$-9.655 \div 0.5$ $0.068 \div -0.01$ $-14.628 \div -4.6$

$\frac{16}{20} \div \left(-\frac{8}{15}\right)$ $\left(-\frac{9}{6}\right) \div \left(-\frac{5}{3}\right)$ $\left(-\frac{3}{5}\right) \div 2\frac{1}{2}$

2.
   a. Write a division sentence in the form of $a \div b$ where the value of a is a number **between** 15.1 and 16.2, and b is a number **between** -4.6 and -3.3.

   Answers will vary.
   $15.3 \div -4.3 = -3.558$

   b. What is the value of the quotient?

   -3.558

3.4.4 GUIDED PRACTICE: FINDING THE QUOTIENTS OF RATIONAL NUMBERS

ENRICHMENT

Notice and Wonder

Before students start to circle the expressions in question 1, consider utilizing a “Notice and Wonder” routine to offer students who still are struggling with sign rules an opportunity to talk through the task to solidify their thinking.

- What do you notice? What is the same? What is different?
- What do you wonder?Can you determine the sign of the quotient without actually solving? How?

3.4.5 YOUR TURN!

TEACHER MOVES

Students may struggle to find the quotients of mixed numbers. Remind students to convert each mixed number to a fraction greater than one before multiplying. Encourage students to apply the rules of dividing integers and pay attention to the signs of the dividend and divisor.

DIFFERENTIATE INSTRUCTION

Consider providing On-Ramp to 7th Grade Accelerated Math videos and practice for students who need additional scaffolding:

- Division of a Unit fraction by a Non-Zero Whole Number
- Division of a Whole Number by a Unit Fraction

3.4.5 Your Turn!

1. Find the value of each expression.

a. $\left(-3\frac{5}{9}\right) \div \left(3\frac{1}{3}\right)$

$-\frac{32}{9} \div \frac{10}{3} = -\frac{32}{9} \times \frac{3}{10}$

$= -\frac{32}{30} = -1\frac{1}{15}$

b. $70.62 \div -160.5$

-0.44

c. $\left(-1\frac{6}{7}\right) \div \left(-5\frac{3}{4}\right)$

$-\frac{13}{7} \div -\frac{23}{4}$

$= -\frac{13}{7} \times -\frac{4}{23}$

$= \frac{52}{161}$

d. $-44.64 \div -2.88$

15.5

e. $8\frac{1}{5} \div \frac{3}{2}$

$\frac{41}{5} \cdot \frac{2}{3} = \frac{82}{15} = 5\frac{7}{15}$

f. $-45.018 \div (-7.32)$

6.15

3.4.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each statement. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can correctly decide the sign of the quotient when dividing rational numbers.

I need help. I can't get started. | I'm getting there. | I understand and have evidence.

Evidence:

Answers will vary.

2. I can fluently divide rational numbers that are fractions.

I need help. I can't get started. | I'm getting there. | I understand and have evidence.

Evidence:

Answers will vary.

3. I can fluently divide rational numbers that are decimals.

I need help. I can't get started. | I'm getting there. | I understand and have evidence.

Evidence:

Answers will vary.

3.4.6 WRAP-UP: REFLECTION

TEACHER MOVES

For students who respond with "I'm getting there," consider prompting them to specifically identify where they are struggling so that it can be addressed.

Compare that with any observations you made during the lesson. What were common mistakes made by your students? What were some roadblocks your students may have had? Consider whether your students may benefit from a day of practice or small-group intervention.